

Matthew Spitzer Brooks

www.mspitzerbrooks.github.io | msbrooks@ucdavis.edu | 781-540-1159

Updated: September 2026

EDUCATION

Ph.D. Agricultural and Resource Economics, University of California, Davis	Expected June 2027
M.S. Agricultural and Resource Economics, University of California, Davis	2024
B.A. Economics and Religion, <i>summa cum laude</i> , Phi Beta Kappa, Middlebury College	2017

REFERENCES

Katrina Jessoe (Chair) Associate Professor Department of Agricultural and Resource Economics University of California, Davis kkjessoe@ucdavis.edu	Travis Lybbert Professor Department of Agricultural and Resource Economics University of California, Davis tlybbert@ucdavis.edu	Jamie Hansen-Lewis Assistant Professor Department of Agricultural and Resource Economics University of California, Davis jhansenlewis@ucdavis.edu
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RESEARCH FIELDS

Environmental Economics, Development Economics, Health Economics

RESEARCH AND WORK IN PROGRESS

The Pollution–Productivity Curve:

Non-Linear Effects and Adaptation in High-Pollution Environments (**Job Market Paper**) [Link to PDF](#)

with Faraz Usmani, Global Energy Alliance for People and Planet

Abstract. This paper provides novel evidence that workers in a high particulate matter air pollution setting partially adapt to chronic exposure, but that adaptation does not offset cumulative harm. Using ball-by-ball data from professional cricket in India, we find that a $10 \mu\text{g m}^{-3}$ increase in same-day $\text{PM}_{2.5}$ reduces performance by about 1 percent. Effects are non-linear, with the largest marginal damages above approximately $50 \mu\text{g m}^{-3}$. Workers with the highest past exposure are roughly 40 percent less sensitive to acute shocks than those with median exposure histories, consistent with adaptation. Nevertheless, chronic exposure degrades performance by more than this adaptation offsets.

Got Goat? The Effects of a Digital Inventory Tool on Livestock Market Outcomes in Rural Nepal

with Travis Lybbert (UC Davis), Conner Mullally (U Florida), Nick Magnan (Colorado State U)

Monetary and Non-monetary Barriers to Accessing Environmental Public Benefit Programs: Experimental Evidence from California

with Shotaro Nakamura (Pennsylvania State U) and Collin Weigel (California Air Resources Board)

Partisan Differences in Adaptation to Wildfire Smoke: Evidence from Indoor Air Quality in California

with Jamie Hansen-Lewis (UC Davis)

Global Spillovers in Agricultural Technology Development

with Ashish Shenoy (UC Davis)

PUBLICATIONS

Peer-reviewed:

- “A Taste for Taxes: Minimizing Distortions Using Political Preferences.” 2019. *Journal of Public Economics*, vol. 180, p. 104055. With Andrea Robbett and Emiliano Huet-Vaughn. [Link](#).

Policy Publications:

- “Does An Innovation’s Reach Reveal Anything About Its Impact? Under The Right Conditions: Possibly.” 2025. Rome, Italy: Standing Panel on Impact Assessment. With Travis Lybbert and Madeleine Walker. [Link to PDF](#).
- “Evaluation of the Liberia Compact’s Mt. Coffee Hydropower Plant Rehabilitation and Capacity Building and Sector Reform: Findings from the Final Round.” 2024. Washington, DC: Mathematica. With Candace Miller, et al. [Link to PDF](#).
- “Liberia Energy Project: Findings from the Final Evaluation of the Pipeline Sub-Activity.” 2024. Washington, DC: Mathematica. With Poonam Ravindranath, Paolo Abarcar, Cullen Seaton, and Candace Miller. [Link to PDF](#).
- “Maternal Health Care Quality Improvement in Rajasthan, India: Insights from a Development Impact Bond Verification Agent.” 2021. Washington, DC: Mathematica. With So O’Neil, Divya Vohra, Shveta Kalyanwala, and Dana Rotz. [Link to PDF](#).
- “Jordan Refugee Livelihoods Development Impact Bond Evaluation Framework.” 2021. Washington, DC: Mathematica. With Evan Borkum, Paolo Abarcar, and Laura Meyer. [Link to PDF](#).
- “Maternal Health Care Quality and Outcomes Under the Utkrisht Impact Bond: Midline Findings and Insights.” 2020. Washington, DC: Mathematica. With So O’Neil, Divya Vohra, Dana Rotz, and Shveta Kalyanwala. [Link to PDF](#).
- “Evaluation of the Liberia Power Compact’s Mt. Coffee Hydropower Plant Rehabilitation and Capacity Building and Sector Reform: Baseline and Interim Findings.” 2020. Washington, DC: Mathematica. With Candace Miller, et al. [Link to PDF](#).

RESEARCH EXPERIENCE

Graduate Student Researcher

University of California, Davis	2024–present
with Dr. Travis Lybbert	2023–2024
with Dr. Ashish Shenoy	

Research Analyst

Mathematica Policy Research, Cambridge, MA	2018–2022
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Policy and Communications Associate

Innovations for Poverty Action, New Haven, CT	2017–2018
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Research Assistant

Princeton University, Princeton, NJ and	
Busara Center for Behavioral Economics, Nairobi, Kenya	
with Dr. Jeremy P. Shapiro	2016–2017

FELLOWSHIPS, HONORS, AND AWARDS

Henry A. Jastro Graduate Research Award, \$3,000, UC Davis	2026
ARE Graduate Summer Research Fellowship, \$3,000, UC Davis	2026
Graduate Studies Travel Award, \$1,500, UC Davis	2026
Henry A. Jastro Graduate Research Award, \$3,000, UC Davis	2025
ARE Graduate Travel Award, UC Davis	2025
SurveyCTO Primary Data Collection Research Grant Honorable Mention	2024
J-PAL Full-RCT grant (collaborator), \$375,000	2024
Henry A. Jastro Graduate Research Award, \$3,000, UC Davis	2024
ARE Graduate Summer Research Fellowship, \$4,000, UC Davis	2024
Provost's Fellowship in the Arts, Humanities and Social Sciences, $\approx$ \$59,000, UC Davis	2022
Phi Beta Kappa, Middlebury College	2017
Highest Honors in Economics, Middlebury College	2017
Highest Honors in Religion, Middlebury College	2017
Religion Departmental Prize, Middlebury College	2017
Dirks Award (Economics senior of high promise), Nominee, Middlebury College	2017
Fulbright Study/Research Grant Nepal, Semi-Finalist	2017
Thomas J. Watson Fellowship, Semi-Finalist	2017
Kellogg Fellow, \$5,000 grant for senior thesis, Middlebury College	2016
College Scholar, Middlebury College	2013-2017

PROFESSIONAL ACTIVITIES

Invited Seminars

Middlebury College, Department of Economics	Jun. 2026
University of Massachusetts - Amherst, Department of Resource Economics	Sep. 2025
Indian Statistical Institute, Delhi	Jul. 2025
Centre for Research on the Economics of Climate, Food, Energy, and Environment	

Conference Presentations

Association for Public Policy Analysis & Management Fall Research Conference (scheduled)	Nov. 2026
Center for Effective Global Action Research Retreat, UC Berkeley (scheduled)	Oct. 2026
Liberal Arts Colleges Development Economics Conference (LACDev; attendee), Wesleyan	Sep. 2026
Agricultural and Applied Economics Association Annual Meeting (4 presentations)	Jul. 2026
World Congress of Environmental and Resource Economists, Portugal	Jun. 2026
Giannini Foundation of Agricultural and Resource Economics Student Conference, UC Berkeley	May 2026
Pacific Conference for Development Economics (PacDev) at UC Davis	Mar. 2026
Occasional Workshop in Environmental & Resource Economics at UC Santa Barbara	Oct. 2025
Agricultural and Applied Economics Association Annual Meeting	Jul. 2025

Referee Service

American Journal of Agricultural Economics, Energy Economics, PLOS One

Departmental Service

Member of Principals of the Community Committee	2025–
Teaching Assistant Coordinator	2025–
Graduate Student Association Representative (Alternate)	2024–
Graduate Student Association Representative	2023-2024

TEACHING EXPERIENCE

Instructor

Applied Economics Research Course (UC Davis, undergraduate)	Summer 2026
A five-week course mentoring students on independent research. Emphasis on designing a research question, working with data, and applying econometric techniques.	

- Prerequisite: ARE 106 — Econometric Theory & Applications.

Teaching Assistant

ARE 133: Introduction to Behavioral Economics (UC Davis, undergraduate)	Fall 2026
ARE 115A: Economic Development (UC Davis, undergraduate)	Winter 2026
ARE 100B: Intermediate Microeconomics (UC Davis, undergraduate)	Winter 2025

Guest Lecturer

ARE 133: Introduction to Behavioral Economics (UC Davis, undergraduate)	April 7, 2026
ARE 176: Environmental Economics (UC Davis, undergraduate)	December 3, 2025

Mentoring

Diversity and Inclusion in Research, Education, and Career Training Program (UC Davis)	
Student Café Graduate Student Coordinator	2026
Graduate Inclusive Education and Mentoring Training (GET) Workshop	2025
Tutor for Undergraduates in Department of Agricultural and Resource Economics Courses	2025

ADDITIONAL

Technical skills:

R, Stata, Python, MATLAB, Julia, Google Earth Engine; L^AT_EX

In-person fieldwork:

Nepal, Kenya, Liberia

Citizenship

U.S. Citizen